

Simpli-Intern AI Executive - What to Learn

Priority: MUST Learn (Class A keywords added to your resume)

- **Data Structures (formal interview-readiness)** — you already have this covered better than most: 6-star CodeChef (2200+ rating) and 1000+ DSA problems solved is a strong, provable signal. The only prep needed is being ready to explain your approach out loud (time/space complexity trade-offs) since interviewers will probe reasoning, not just correctness.
- **Machine Learning (ML) fundamentals** — as with prior applications, your background is applied GenAI (LLM APIs, RAG, agents) rather than classical ML. Learn the core vocabulary: supervised vs. unsupervised learning, train/test split, overfitting, bias-variance trade-off. Andrew Ng's "Machine Learning Specialization" intro module (audit free on Coursera) covers the essential 20%.
- **Natural Language Processing (NLP)** — you already do NLP-adjacent work (RAG semantic search, intent handling in Echo), so this is largely a labeling exercise: learn the formal vocabulary (tokenization, embeddings, classification, NER) to describe what you've already built.
- **Deep Learning** — know the basic building blocks of the neural networks that power the LLMs you already use daily (layers, weights, backpropagation, transformers at a conceptual level). 3Blue1Brown's neural network video series is the fastest way to get conversational.
- **Python (deepen for ML context)** — you know Python at a basic/backend level; the delta for this JD is data-science libraries: pandas, numpy, and a basic scikit-learn fit/predict workflow. A few hours with a scikit-learn quickstart covers most interview questions.

Good to Know (Class B — strengthens your fit for this specific JD)

- **Statistical modeling** — basic statistics (mean/variance, hypothesis testing, correlation vs. causation) is worth a light review; it's the kind of question that comes up in an "insights from data" interview.
- **Cloud platforms (AWS/Azure)** — you already have real AWS Secrets Manager experience (Echo); Azure is more of a "different vendor, same concepts" situation. Skim Azure's equivalents to what you know on AWS (Key Vault vs. Secrets Manager, App Service vs. deployment) so you can speak to both.
- **Omnichannel speech/voice for conversational AI** — you already have a genuine hook here: Echo integrates Vapi for voice-assistant support. Be ready to describe that integration in interview detail (webhook handling, live transcripts) since it's your strongest direct match to this specific requirement.

How this maps to your background

This role wants applied Generative AI skills (which you have in depth via Echo and Vibe — RAG, LLM function calling, prompt engineering, and even voice/conversational AI through Vapi) layered on top of classical ML/DL/NLP theory and statistics (which you don't have formal exposure to yet). Your data-structures/competitive-programming background is a genuine differentiator most candidates won't have. Prioritize ML/DL/NLP vocabulary first, keep the CodeChef/DSA story polished for the interview, and lean on the Echo voice-AI integration as your standout talking point for the "conversational AI" requirement.